# Executable Composition and Post-Composition Event Admissibility in Partial Ternary Systems

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## Abstract

We study partial transformations over finite ternary configurations in which the value U denotes traceably established non-closure after the admissible resolution paths available within a declared domain and horizon have been exhausted. U is therefore neither a probability nor an unknown, pending, or inconclusive truth value. We define admissible events by typed domains, legible input and output configurations, declared support, exterior control, observational compatibility, and a constitutive non-triviality condition. The main result separates executable composition from event formation: a typed composite transformation may be well defined on a nonempty effective domain yet fail to constitute an admissible event because its net effect is null for every compatible observable. An explicit finite counterexample establishes this failure of closure under composition. We then distinguish pairwise executability from joint realizability by requiring a common witness for an entire event sequence, and show that pairwise executable composites need not admit such a witness. For realized chains we obtain a telescoping identity for compatible observables. Finally, under append-only ternary persistence, the number of U-valued coordinates is a strictly decreasing rank for a restricted class of strict-enablement chains, yielding acyclicity and a finite length bound. The framework is event-driven and requires neither a primitive time parameter nor probabilistic semantics.

**Keywords:** partial maps; event semantics; realizability; ternary systems; non-closure; concurrency

## 1. Introduction

Formal models of computation routinely separate states from transitions, and several mature theories describe event occurrence, partial maps, concurrent histories, observable behavior, and incomplete specifications [1-9]. What is less commonly separated is the following pair of questions:

1. Is a composite transformation executable on at least one input?
2. If it is executable, does the composite itself still qualify as an event of the system?

The distinction is easy to miss because many familiar formalisms build event or transition status into the primitive relation. Event structures describe occurrences and their causal or conflict relations [1,2]. Event automata make the relationship even more explicit: each state change coincides with the occurrence of an event [3]. Categories of partial maps and restriction categories formalize partiality and composition without adding an independent criterion for whether a resulting composite is still an event [4,5]. Modal and three-valued transition systems address incomplete or abstract state spaces [6-8], while process calculi may include silent transitions that remain transitions even when they are hidden from observation [9].

This paper studies a different architecture. A transformation can be typed, partially defined, executable, and compositionally well formed without thereby acquiring or preserving event status. Event status is constitutive: after composition, the candidate composite must again satisfy the conditions that define an admissible event. One of these conditions requires nonzero net variation for at least one compatible observable. Consequently, two individually admissible events may compose to the identity on their effective domain and thereby cease, as a composite, to constitute an admissible event.

The framework uses finite configurations over the alphabet

$\Sigma = \{0, 1, U\}$.

The symbol U does not mean unknown, transition, cancellation, probability, or an inconclusive intermediate verdict. It denotes non-closure that is legitimate only after the admissible resolution paths available in the declared domain and horizon have been traceably exhausted without sufficient basis for closure to 0 or 1. This semantic restriction matters because ternary formalisms already exist for very different purposes. In three-valued model checking, for example, the third value represents information lost through abstraction or an outcome that remains unknown at the current abstraction level [6-8]. Pratt's triadic higher-dimensional account uses a third value to denote transition and later adds a distinct cancellation value [10]. Neither interpretation is adopted here.

The paper makes four contributions.

First, it defines the effective domain of a finite sequence of typed partial reevaluations and separates the resulting composite map from the event-admissibility test. The central theorem gives a finite counterexample in which two admissible events are jointly executable but their composite fails the observational non-triviality condition and is therefore not an admissible event.

Second, it introduces joint realizability through a common witness. Pairwise executability is shown not to imply joint realizability, even when every individual step is an admissible event.

Third, it derives a telescoping identity for compatible observables along realized chains. This result is elementary as an algebraic identity; its role is to make explicit which accumulation statement is justified once a common realized path has been established.

Fourth, it studies an append-only ternary regime in which a coordinate that has been closed to 0 or 1 cannot return to U within the same trajectory. For strict-enablement chains in which each step closes at least one necessary U-valued coordinate, the number of U-valued coordinates is a strictly decreasing rank. The generic use of ranking functions for termination is classical [11]; the contribution here is the specific rank induced by ternary non-closure and append-only persistence within the present event architecture.

The formal development is self-contained. It is an external reconstruction of results developed in the Sistema Vectorial SV line of work, whose earlier public stages introduced the ternary configuration space, event-driven reevaluation, admissible events, event chains, and append-only persistence between March and April 2026 [12-18]. The complete Spanish preprint from which the present article is reconstructed was published on 17 August 2026 [19]. The present article does not require the reader to adopt that prior notation or consult those publications in order to verify the definitions and proofs below.

The remainder of the paper is organized as follows. Section 2 states the methodological scope and provenance. Section 3 positions the framework against the closest external literature. Section 4 defines ternary configurations, structural horizons, observables, and admissible events. Sections 5 and 6 establish the separation between composition, event admissibility, and joint realizability. Section 7 treats observable accumulation. Section 8 derives append-only closure ranks and acyclicity. Section 9 addresses change of horizon and recurrence. Section 10 gives a finite verification procedure and its computational interpretation. Section 11 discusses scope and limitations, and Section 12 concludes.

## 2. Methodological scope, provenance, and source verification

### 2.1. Mathematical scope

The aim is not to provide a universal ontology of events. The paper studies a typed mathematical regime with four deliberately limited ingredients: finite ternary configurations, partial reevaluation maps, a constitutive event-admissibility test, and append-only persistence for closed coordinates. No metric, probability measure, global identity event, total composition law, or continuous time parameter is assumed.

The index of a finite event sequence records order only. It is not interpreted as physical or canonical time. External clocks may be attached to applications, but no theorem below depends on elapsed time. Likewise, no probability distribution is attached to U. The semantic meaning of U is non-closure under a declared resolution protocol, not uncertainty quantified by frequency or belief.

The core theorems are first proved in a fixed-horizon regime because this is the weakest setting in which the distinction between executable composition and admissible event formation can be stated without introducing unnecessary transport machinery. Section 9 then records the additional requirement needed when horizons change: cross-horizon identity must be carried by an explicitly declared transport rather than presumed from notation or visual similarity.

### 2.2. Provenance

The present formalism has a documented development history. Ternary configurations and the basic algebraic semantics were public by 9 March 2026 [12]. Event-driven reevaluation and append-only trajectories were public by 11 March [14]. The semantic status of U as non-probabilistic non-closure was published on 14 March [13]. A formal theory of admissible events appeared on 22 March [16], chains and accumulation on 25 March [17], and append-only preservation and branching on 30 March [18]. The March event theory explicitly left partial event composition open [16]. The present paper resolves that question in the limited sense established in Sections 5 and 6: composition of the underlying reevaluations is separated from revalidation of the composite candidate as an event. The consolidated Spanish preprint *Dinámica del Suceso - Sistema Vectorial SV* was published on 17 August 2026 with DOI 10.21428/39829d0b.8ea18396 [19].

This chronology is reported for provenance, not as evidence of dependence by later work. In particular, Lee's *Guarded Realization Semantics*, first submitted to arXiv on 26 July 2026, is a recent and technically relevant neighbor in pathwise domains, realization-sensitive structure, and partial-map transport [20]. No claim of dependence is made in either direction. The comparison in Section 3 is structural.

### 2.3. Literature search and AI-assisted research support

The literature search was conducted by tracing exact titles, DOIs, publisher records, and public preprints for event structures, partial maps, three-valued transition systems, cancellation, dynamic causality, termination ranks, and realization semantics. Candidate references surfaced by search tools were checked against publisher or primary-source records before inclusion.

OpenAI ChatGPT (GPT-5.6 Sol), Grok 4.5 (xAI), and DeepSeek-V4-Pro (DeepSeek AI) were used in a supporting role for literature discovery, adversarial consistency checks, structural review, and assistance with formalization. Their outputs were not treated as mathematical authority or as substitutes for source verification. The formal declaration required by the journal appears before the references.

## 3. Related work and positioning

### 3.1. Event structures and event automata

The event-structure tradition models concurrent computation through events, causal dependency, conflict, and configurations representing consistent histories [1,2]. Later work allows causal dependencies themselves to change during execution [21]. These models are central neighbors because they make event occurrence and history mathematically explicit.

Event automata provide an especially useful contrast. Pinna and Poigné define states that retain information about the events that have happened and impose the characteristic condition that every state change coincides with the happening of an event [3]. The framework developed here does not impose that identification. A reevaluation may be a well-typed executable transformation while failing to constitute an admissible event. More strongly, a composite of individually admissible events may be executable and yet fail the event-admissibility test after composition.

This is not a claim that event structures lack observability, consistency conditions, or execution semantics. They have all three in substantial forms. The narrower claim is that, in the literature examined, we have not found the same two-stage architecture in which a partial composite is first established as executable and then independently revalidated as an event by a constitutive observational non-triviality condition.

### 3.2. Partial maps and restriction categories

Partial composition is classical. Restriction categories provide an abstract categorical account of partial maps in which each arrow carries a restriction idempotent encoding its domain of definition [4], and the subsequent theory develops partial-map classification in this setting [5]. The effective-domain construction in Section 5 therefore makes no claim to invent partial composition or its associativity.

The distinction lies elsewhere. In a partial-map formalism, once a composite arrow exists, it remains an arrow of the category. Here the composite partial map and the candidate composite event are different mathematical objects. The map may exist while the event candidate fails one of the constitutive admissibility conditions. The central counterexample exploits precisely this separation.

### 3.3. Silent transitions, three-valued abstractions, and cancellation

Process-algebraic models commonly distinguish observable from unobservable actions; a silent action remains a transition of the operational system even when it is abstracted from external observation [9]. The present non-triviality condition does something different. It does not hide an already constituted event. It decides whether the candidate composite belongs to the event class at all.

Three-valued model checking and modal transition systems are also close in notation but not in semantics. Modal transition systems support partial specifications and three-valued program analyzes [6]. Three-valued abstraction uses true, false, and unknown, where unknown represents information lost through abstraction or insufficient precision for a definite model-checking outcome [7,8]. U in the present paper is not such an unknown value. A missing computation, an unrefined abstraction, or an observation that has not yet been completed does not by itself justify U. U is reserved for traceably established non-closure after the declared admissible resolution paths have been exhausted.

A further important neighbor is Pratt's treatment of transition and cancellation in higher-dimensional automata [10]. There a third value denotes transition, and a fourth value is introduced for cancellation. The present U denotes neither. It does not say that a coordinate is currently changing, nor that an event has been cancelled. It records a non-closure status under a declared resolution horizon.

These distinctions are summarized in Table 1.

**Table 1. Positioning of U and post-composition admissibility against selected neighbors.**

| Framework | Third/non-Boolean status | Meaning of that status | Partial execution | Composite may exist yet fail event status by net observational triviality |
|---|---|---|---|---|
| Three-valued model checking [6-8] | yes | unknown / abstraction loss / insufficient precision | model-dependent | not found in the examined formulation |
| Pratt's triadic HDA [10] | yes | transition; separate cancellation value | yes, in its own semantics | different mechanism |
| Event automata [3] | not defined as U | event occurrence tied to state change | constructions on automata | state change is itself event occurrence |

| Framework | Third/non-Boolean status | Meaning of that status | Partial execution | Composite may exist yet fail event status by net observational triviality |
|---|---|---|---|---|
| Restriction categories [4,5] | no U semantics | domain of partial maps | central | composite remains a morphism when defined |
| Present framework | yes | traceably established non-closure | central | yes; this is Theorem 5.2 |

### 3.4. Realization-sensitive paths and recent work

The distinction between visible behavior and the structure of a realization is not new. Event structures preserve histories, and several semantics distinguish paths or realizations that lead to similar external results. Recent work by Lee develops a guarded realization semantics in which distinct proofs, programs, formulas, or rewrite paths may have the same observable behavior while differing in occurrence structure, interfaces, and transformation history [20]. It also studies the greatest subobject transported intact through a finite rewrite path and develops pathwise certificates.

This is close to the common-witness and effective-domain perspective of Sections 5 and 6. Accordingly, joint realizability is not presented here as an isolated claim of international novelty. The notions are nevertheless not identical. Lee's pathwise certificates and greatest transported subobjects carry quantitative and structural information through rewrite paths [20]; the common witness used here is simply an initial configuration $x_0 \in D_{1\ldots m}$ that certifies execution of one whole path. It is neither a transported subobject nor a quantitative certificate. Conversely, the present paper does not reproduce Lee's richer certificate machinery. The central difference for the result studied here is the post-composition event-admissibility layer: the existence of a pathwise composite is not sufficient for event status, because the composite is independently tested for constitutive non-triviality.

### 3.5. Ranking arguments

Termination proofs by well-founded orderings and ranking functions are classical [11]. The rank introduced in Section 8 should be read in that tradition. Its specific content is that append-only persistence of ternary closure makes

$u(S) = |\{i : S(i) = U\}|$

a natural rank for strict-enablement chains that close at least one necessary U-coordinate per step. The generic proof pattern is established mathematics; the particular rank is a consequence of the non-closure semantics and persistence rule used here.

### 3.6. Novelty claim after comparison

The contribution is therefore not the isolated invention of partial composition, a third value, pairwise-versus-global path distinctions, a telescoping sum, or a decreasing termination measure. In the literature examined, we have not found an equivalent event framework combining all of the following:

- a ternary value U denoting traceably established non-closure rather than unknown, probability, transition, or cancellation;
- event-driven dynamics without time or probability as primitives;
- partial executable composition over an effective domain;
- independent post-composition revalidation of event admissibility by observational non-triviality;
- joint realizability through a common witness; and

- append-only preservation of closed coordinates, yielding a finite closure rank for strict-enablement chains.

The strongest result is the fourth item: a typed partial composite may be executable on a nonempty effective domain yet fail to constitute an event solely because its net observable variation is zero.

# 4. Formal setting

## 4.1. Ternary configurations and non-closure

Let

$\Sigma = \{0, 1, U\}$.

For a finite index set I, a ternary configuration is a map

$S : I \to \Sigma$.

When $I = \{1, \ldots, n\}$, we write $S = (s_1, \ldots, s_n) \in \Sigma^n$.

The values 0 and 1 are closed values. Their domain-specific interpretations are not fixed globally by the mathematics; they may encode, for example, verified non-incidence and incidence, rejection and acceptance, or other binary closures specified by an application.

The value U has a stricter semantic condition.

**Definition 4.1 (legitimate non-closure).** A coordinate S(i) may be assigned U only when the admissible resolution paths declared for coordinate i in the current domain and horizon have been traceably exhausted without sufficient basis for closure to either 0 or 1.

Definition 4.1 has three immediate negative consequences. U is not justified merely because a computation has not yet been run. It is not a probability or confidence score. It is not synonymous with an abstract unknown that may disappear simply by refining a representation. A concrete application may later acquire new admissible information and close U to 0 or 1, but the original U records the non-closure that was justified under the earlier declared horizon.

The resolution trace that justifies U is evidence associated with the assignment; it is not an additional element of Σ. Its internal certificate syntax is intentionally abstracted here: the results below use traceable exhaustion as a certified premise attached to the U assignment, not as a fourth semantic value. Thus Σ remains exactly ternary.

## 4.2. Structural horizons

A structural horizon is a tuple

$H = (I\_H, \preccurlyeq\_H, X\_H, A\_H)$,

where I_H is a finite set of positions, ≼_H is a declared internal relation on positions, X_H ⊆ Σ^{I_H} is the set of legible configurations, and A_H is a family of real-valued observables $F : X\_H \to \mathbb{R}$.

The relation ≼_H is intentionally left weak. None of the results below requires it to be total, metric, temporal, or causal. Applications may strengthen it.

For two horizons H and H′, let

$CObs(H,H') \subseteq A\_H \times A\_H'$

be the declared compatibility relation between observables. A pair (F,F′) ∈ CObs(H,H′) means that the two readings are licensed for comparison across the corresponding horizons. For the fixed-horizon counterexamples of Sections 5 and 6 and the rank regime of Section 8, observable compatibility is taken to be diagonal: CObs(H,H) = {(F,F) : F ∈

$A_H\}$. Applications may declare richer cross-horizon compatibility relations, but those are not used in the cancellation theorem.

### 4.3. Admissible events

An event candidate is a quadruple

$e = (H, H', \sigma, R_e)$,

where $\sigma \subseteq I_H$ is a declared support and $R_e$ is a partial map with nonempty domain $D_e \subseteq X_H$ and codomain $X_{H'}$.

The candidate is an **admissible event** when the following six conditions hold.

**(A1) Typed support.** $\sigma \subseteq I_H$.

**(A2) Nonempty typed domain.** $D_e \neq \emptyset$ and $R_e : D_e \to X_{H'}$.

**(A3) Legibility.** Every $x \in D_e$ is legible in H and every $R_e(x)$ is legible in H′.

**(A4) Constitutive observational non-triviality.** There exist $x \in D_e$ and $(F,F') \in \mathrm{CObs}(H,H')$ such that

$F'(R_e(x)) - F(x) \neq 0$.

**(A5) Exterior control.** There exist $J_e \subseteq I_H \setminus \sigma$ and a declared partial transport $\theta_e : J_e \to I_{H'}$ that records how the selected exterior positions are related to the output horizon.

**(A6) Observational compatibility.** Every cross-horizon observable comparison invoked to establish admissibility belongs to CObs(H,H′).

Condition (A4) is constitutive rather than merely descriptive. It excludes a transformation from event status when the candidate produces no nonzero net difference for any compatible observable on any input in its domain. A4 is qualitative, not a universal magnitude threshold: the application declares $A_H$ and the compatibility relation that determine which observables are admissible for the stated horizon. If an application requires a threshold or a stronger relevance criterion, that requirement must be encoded in its declared observable semantics rather than inferred from A4.

This point should not be confused with observability in labeled transition systems. An operationally silent transition may still be a transition. Here the question is whether the candidate transformation belongs to the event class at all.

### 4.4. Event sequences and trajectories

Let

$e_k = (H_{k-1}, H_k, \sigma_k, R_k)$, $k = 1, \ldots, m$,

be admissible events whose arrival and departure horizons match consecutively. A realized trajectory is a sequence

$x_0 \xrightarrow{e_1} x_1 \xrightarrow{e_2} \ldots \xrightarrow{e_m} x_m$

with $x_0 \in D_1$ and $x_k = R_k(x_{k-1})$ for each k.

The subscript k records succession. No duration or time metric is implied.

## 5. Executable composition and post-composition admissibility

### 5.1. Effective domain

A sequence of partial maps can be typed without being jointly executable on any common input. We therefore define the domain of the whole sequence explicitly.

**Definition 5.1 (effective domain).** For $e_1, \ldots, e_m$ as above, the effective domain is

$D_{1\ldots m} = \{x \in D_1 : R_1(x) \in D_2, R_2(R_1(x)) \in D_3, \ldots, (R_{m-1} \circ \ldots \circ R_1)(x) \in D_m\}$.

If $D_{1\ldots m} \neq \emptyset$, the sequence is executable on at least one common initial configuration.

The associated composite reevaluation is

$R_{1\ldots m} = R_m \circ \ldots \circ R_1$, restricted to $D_{1\ldots m}$.

This is standard partial-function composition. In particular, on the common domain where every application is defined, composition is associative.

**Proposition 5.1 (functional associativity).** For any three consecutively typed reevaluations,

$R_3 \circ (R_2 \circ R_1) = (R_3 \circ R_2) \circ R_1$

on their common effective domain.

**Proof.** This is ordinary associativity of function composition restricted to the set on which all three successive applications are defined. (q.e.d.)

The proposition concerns maps, not event admissibility.

### 5.2. Executability does not imply event admissibility

Suppose $D_{1\ldots m} \neq \emptyset$. A natural candidate composite event has the form

$e_{1\ldots m} = (H_0, H_m, \sigma_{1\ldots m}, R_{1\ldots m})$,

where $\sigma_{1\ldots m}$ is any support declaration satisfying the relevant typing conditions. The existence of $R_{1\ldots m}$ does not imply that this candidate is an admissible event.

**Theorem 5.2 (failure of closure of event admissibility under executable composition).** There exist admissible events $e_1$ and $e_2$ such that $D_{1\ldots 2} \neq \emptyset$ and $R_{1\ldots 2}$ is well defined, but the composite candidate $e_{1\ldots 2}$ is not an admissible event.

**Proof.** Work in the fixed horizon H with $I_H = \{1,\ldots,9\}$ and assume that $X_H$ contains

$S_0 = (0,0,0,0,0,0,0,0,0)$,

$S_1 = (1,0,0,0,0,0,0,0,0)$.

Let $D_1 = \{S_0\}$ with $R_1(S_0) = S_1$, and let $D_2 = \{S_1\}$ with $R_2(S_1) = S_0$. For both events choose $\sigma = \{1\}$. Let $J = \{2,\ldots,9\}$ and use $\theta = id_J$ as exterior control. Take the fixed-horizon compatibility relation to be the diagonal relation specified in Section 4.2. Let $F \in A_H$ be

$F(S) = |\{i : S(i) = 1\}|$.

Then $F(S_1) - F(S_0) = 1$ and $F(S_0) - F(S_1) = -1$. Hence each individual transformation has nonzero observable variation. The domains are nonempty, all states are legible in H, the support and exterior control are well typed, and fixed-horizon observational comparison is compatible. Thus both $e_1$ and $e_2$ satisfy (A1)-(A6).

However,

$D_{1\ldots 2} = \{S_0\}$

and

$R_{1\ldots 2}(S_0) = S_0$.

Therefore, for every compatible observable $G \in A\_H$,

$G(R_{1\ldots 2}(S_0)) - G(S_0) = 0$.

The composite candidate fails (A4) and is not an admissible event. (q.e.d.)

The theorem establishes the central separation:

**executable composite transformation ≠ admissible composite event.**

The failure is not a typing failure, a missing domain, or an observationally hidden transition. The composite exists as a partial map and is executable on a nonempty domain. It fails only at the independent constitutive test for event admissibility.

### 5.3. Post-composition revalidation

The preceding theorem yields an immediate criterion.

**Corollary 5.3 (post-composition revalidation).** A well-defined composite reevaluation constitutes an admissible composite event if and only if the resulting event candidate satisfies (A1)-(A6) again.

Thus two checks are logically independent:

1. **Execution check:** $D_{1\ldots m} \neq \emptyset$.
2. **Event check:** the candidate composite satisfies (A1)-(A6).

No algebraic completion is assumed to convert the first into the second.

### 5.4. The identity map is not an event by default

The identity map is neutral for function composition:

$R \circ \mathrm{id} = R = \mathrm{id} \circ R$.

Yet for every compatible observable F,

$F(\mathrm{id}(x)) - F(x) = 0$.

Therefore the identity map does not, by itself, constitute an admissible event because it fails (A4). This does not forbid local neutral constructions in other specialized formalisms; it only shows that functional identity does not automatically become an event in the present general regime.

## 6. Joint realizability

### 6.1. Realizable segments

Pairwise executable composition is weaker than execution of an entire sequence by one initial configuration.

**Definition 6.1 (realizable segment).** A finite event sequence $(e_1,\ldots,e\_m)$ is jointly realizable when $D_{1\ldots m} \neq \emptyset$. Any $x_0 \in D_{1\ldots m}$ is a common witness, and it induces the realized path $x\_k = R\_k(x\_\{k-1\})$.

The word "jointly" is important. Each adjacent pair may have some witness even when no single witness traverses the whole sequence.

## 6.2. Pairwise executability does not imply joint realizability

**Proposition 6.2.** There exist three admissible events $e_1$, $e_2$, $e_3$ such that $D_{1\ldots2} \neq \emptyset$ and $D_{2\ldots3} \neq \emptyset$, while $D_{1\ldots3} = \emptyset$.

**Proof.** Work again in a fixed horizon H over nine coordinates and assume that X_H contains

a = (0,0,0,0,0,0,0,0,0),

b = (1,0,0,0,0,0,0,0,0),

c = (0,1,0,0,0,0,0,0,0),

d = (1,1,0,0,0,0,0,0,0),

e = (0,0,1,0,0,0,0,0,0).

Define

$D_1 = \{a\}$, $R_1(a) = b$, $\sigma_1 = \{1\}$,

$D_2 = \{b,c\}$, $R_2(b) = d$, $R_2(c) = e$, $\sigma_2 = \{2,3\}$,

$D_3 = \{e\}$, $R_3(e) = a$, $\sigma_3 = \{3\}$.

Use the diagonal fixed-horizon compatibility relation of Section 4.2 and let F(S) count the coordinates equal to 1. For each k, set J_k = I_H \ σ_k and choose θ_k = id_{J_k} as exterior control. The three domains are nonempty; every listed input and output belongs to X_H; each support is a subset of I_H; and every R_k maps its declared domain into X_H. Thus (A1)-(A3) and (A5)-(A6) hold. Condition (A4) is witnessed by a for $e_1$, by b for $e_2$, and by e for $e_3$, since the corresponding F-variations are 1, 1, and -1. Hence $e_1$, $e_2$, and $e_3$ are admissible events.

The pair $(e_1,e_2)$ is executable from a because a → b → d. The pair $(e_2,e_3)$ is executable from c because c → e and e ∈ $D_3$. But the only trajectory starting from $D_1$ is

a → b → d,

and d ∉ $D_3$. Hence $D_{1\ldots3} = \emptyset$. (q.e.d.)

The proposition is a basic fact about partial domains, not an isolated novelty claim. Its role is to prevent a local composability condition from being silently promoted to a global path claim.

## 6.3. Realizable chains

For this paper, a **chain** is a consecutively typed event sequence satisfying the local conditions needed for compositional reading: each adjacent pair is executable, every cross-horizon identification used by the chain is licensed by declared transport data, every observable comparison is licensed by CObs, and an accumulation rule is specified. These are local conditions; they need not supply a single global witness for the whole sequence.

**Definition 6.3 (realizable chain).** A chain is realizable when it also has a common witness for the entire path.

In a fixed horizon this is exactly $D_{1\ldots m} \neq \emptyset$. Under horizon changes, Section 9 requires the witness to be preserved through the declared transport.

A realizable chain still need not be an admissible composite event. The chain may exhibit nontrivial intermediate changes and yet return to an observationally identical endpoint, as in Theorem 5.2. Realizability concerns the existence of the path; event admissibility concerns the status of the composite transformation.

## 7. Observable accumulation on realized paths

Let $x_0 \to x_1 \to \dots \to x_m$ be a realized chain. For each horizon $H_k$, let $F_k \in A_{H_k}$ be a compatible reading along the path, with each consecutive pair licensed by the declared observational compatibility relation.

**Theorem 7.1 (telescoping observable variation).**

$$F_m(x_m) - F_0(x_0) = \Sigma_{k=1}^m [F_k(x_k) - F_{k-1}(x_{k-1})].$$

**Proof.** Expanding the right-hand side cancels every intermediate term once with positive and once with negative sign, leaving only $F_m(x_m) - F_0(x_0)$. (q.e.d.)

The theorem is algebraically elementary. Its purpose is structural: the equality is justified only after a realized path and compatible observable readings have been fixed. It does not establish that the entire chain is one admissible event. If the endpoint difference is zero for every compatible observable, the composite fails (A4) even though the intermediate sum contains nonzero terms.

This is exactly what happens in the cancellation counterexample of Theorem 5.2:

$$1 + (-1) = 0.$$

Intermediate eventhood and composite eventhood are therefore distinct.

## 8. Append-only closure and finite strict-enablement rank

### 8.1. Append-only ternary persistence

We now impose one additional rule on trajectories in a fixed horizon.

**Definition 8.1 (append-only closure).** A ternary trajectory $S_0, S_1, \dots$ is append-only with respect to closed coordinates if, whenever $S_k(i) \in \{0,1\}$, then

$$S_l(i) = S_k(i)$$

for every $l > k$ in the same trajectory.

Thus a coordinate that has closed to 0 or 1 cannot later return to U or switch to the opposite closed value within that trajectory. Branching is permitted: two trajectories may share a common prefix and later close previously open coordinates differently. What is forbidden is retrospective rewriting of the already constituted prefix.

This rule is not a probabilistic convergence assumption and does not use time. It is a persistence condition on the ordered record of a trajectory.

### 8.2. Closure rank

For a finite configuration S, define

$$u(S) = |\{i \in I_H : S(i) = U\}|.$$

The function u takes values in the nonnegative integers.

**Definition 8.2 (strict U-enablement chain).** A realizable append-only chain

$$S_0 \xrightarrow{e_1} S_1 \xrightarrow{e_2} \dots \xrightarrow{e_m} S_m$$

is strictly U-enabling if every step closes at least one coordinate that had value U and, for each $k < m$, at least one closure effected by $e_k$ is required to enable $e_{k+1}$.

The second clause limits the theorem to a genuine enablement regime rather than every arbitrary event sequence, while the first ensures strict decrease at every step, including the terminal one.

**Theorem 8.3 (strict decrease and acyclicity).** In every strict U-enablement chain,

$$u(S_{k+1}) < u(S_k)$$

for every k. Consequently, the chain cannot contain a cycle.

**Proof.** Each step closes at least one U-valued coordinate. By append-only closure, no previously closed coordinate can return to U in the same trajectory. Hence the set of U-valued coordinates strictly decreases at every step, so its cardinality u(S) strictly decreases in $\mathbb{N}$. A cycle would require a return to a previous configuration and therefore to a previous value of u, which is impossible under strict decrease. (q.e.d.)

**Corollary 8.4 (finite bound).** If $|I_H| = n$, then

$$m \leq u(S_0) \leq n.$$

In the canonical nine-coordinate case, $m \leq 9$.

The theorem does not assert that every event relation is acyclic. Cycles may exist outside the strict U-enablement regime, and different witnesses may support different pairwise relations without forming one jointly realizable trajectory. The result is intentionally local to the stated hypotheses.

# 9. Horizon changes, transport, and recurrence

## 9.1. No default identity across horizons

When H changes to H′, the index sets, legible configuration spaces, or observable families may change. Equality of labels is therefore insufficient to identify positions across horizons.

Let $T_{H \to H'}$ be a declared partial transport on the subset of positions for which cross-horizon identification is justified. This generic transport should not be confused with the event-specific exterior-control map $\theta_e$ in (A5), although an application may relate them. No uniqueness, totality, or general composition law for transports is assumed here. The only principle needed for the present results is negative:

**No transport, no cross-horizon identity claim.**

The same applies to observables. A numerical expression in $A_H$ and a formally similar expression in $A_{H'}$ are comparable only when the compatibility relation CObs(H,H′) declares them comparable.

This restriction avoids silently importing continuity. A chain may cross horizons, but each inheritance or comparison used in a proof must be carried by declared transport data. If several horizon changes occur, every cross-horizon identification actually used in the argument must be justified; a general coherence theorem for composed transports is outside the scope of this paper.

## 9.2. Recurrence without event identity

The same visible configuration can arise in distinct evaluations or trajectories. Formally, one may have configurations $S_a$ and $S_b$ with

$$S_a = S_b$$

while the events leading to them or the realized paths differ.

This does not contradict append-only closure. Append-only persistence constrains one trajectory. It does not identify distinct trajectories merely because their current visible configurations coincide.

The distinction is important computationally. A state-vector equality test cannot reconstruct path provenance. Systems that need event identity or path-sensitive behavior must retain trajectory information in addition to the visible configuration. This point is consistent with the wider literature on histories and realization-sensitive semantics [1-3,20].

## 10. Finite classification procedure and computational interpretation

The preceding distinctions give a direct decision procedure for finite, explicitly represented systems. This procedure is not an algorithmic novelty claim; its role is to make the semantic separation computationally explicit.

Assume that every $D_k$ is finite and enumerable, each $R_k$ can be evaluated on its declared domain, support and exterior-control data are explicit, and the relevant compatible observable family is finite. Given $e_1,\dots,e_m$, propagate each $x \in D_1$ through $R_1,\dots,R_m$ and discard it when a required application is undefined. The surviving initial configurations are exactly $D_{1\dots m}$. If none survive, the sequence is not jointly executable. Otherwise construct $R_{1\dots m}$ on $D_{1\dots m}$ and revalidate the candidate composite against (A1)-(A6). The outcome is therefore one of three meta-level classifications: no jointly executable composite, executable composite that is not an admissible event, or admissible composite event. These classifications are not the three values of $\Sigma$.

With explicit finite representations and constant-time map evaluation, a direct scan uses at most $m|D_1|$ partial-map evaluations to compute the effective domain; if q compatible observable pairs are then inspected for (A4), the core execution and observational checks are bounded by $O(m|D_1| + q|D_{1\dots m}|)$, apart from representation-dependent typing and transport costs. The bound is deliberately elementary.

The computational point is the distinction of outcomes. A workflow engine, symbolic executor, rule system, or transition service may establish that a sequence of partial operations is executable and still need a second decision: whether its net effect is nontrivial under the declared observable semantics and therefore constitutes an event. Undefined composition and executable observational triviality are different cases and should not be collapsed into one generic failure state. Symbolic or infinite-state realizations may replace explicit enumeration by SAT/SMT, automata-theoretic, categorical, or abstract-interpretation methods; those extensions do not alter the finite results proved here.

## 11. Discussion and limitations

### 11.1. What the main theorem does and does not claim

Theorem 5.2 does not claim that composition of partial functions is novel. It does not claim that observational equivalence is novel, nor that cancellation as a mathematical phenomenon is novel. Pratt's work explicitly studies transition and cancellation in concurrency [10]. The theorem states something narrower: event admissibility is not closed under executable composition when admissibility includes an independently rechecked net observational non-triviality condition.

The counterexample is deliberately finite and minimal in conceptual structure. Two admissible events change one coordinate in opposite directions. Each is nontrivial; the composite is the identity on its effective domain and fails the event test. Exact cancellation is chosen deliberately because it makes failure of (A4) independent of the separating power of the observable family: with diagonal fixed-horizon compatibility, every observable has the same value before and after the composite. A non-identity endpoint would instead require additional assumptions about observational indistinguishability. The proof therefore does not depend on stochasticity, continuous dynamics, time, or asymptotic arguments.

### 11.2. U is not a generic third truth value

Three-valued formalisms can share the same alphabet size while having incompatible semantics. In abstraction-based model checking, unknown records lack of precision in the abstract model [6-8]. In Pratt's triadic account, the third value records transition [10]. Here U records non-closure after admissible resolution has been exhausted.

The distinction is operational. If a computation remains to be performed, the correct response is to perform it; pending work is not U. If an abstraction can be refined to recover information, the current abstraction's unknown value does not thereby establish U. U is justified only by the declared exhaustion condition of Definition 4.1.

No theorem in this paper requires U to be probabilistic, fuzzy, epistemic, or temporal. Nor does the paper claim that no previous formalism has ever used a third symbol under a non-probabilistic interpretation. The claim is semantic specificity, not cardinal novelty.

### 11.3. No primitive time

The event index k orders a trajectory but does not measure duration. This is compatible with many untimed computational models and is not by itself a novelty claim. The point is that none of the definitions or proofs needs elapsed time as the primitive cause of change. Applications that require deadlines, timed guards, stochastic rates, or physical durations may add them externally, provided they do not replace the event-admissibility conditions being studied.

### 11.4. Realizability is necessary but not sufficient

A common witness is necessary to claim a realized path, but it does not imply that the entire path forms one event. The realized chain and the composite event answer different questions. The first asks whether one input can traverse the sequence. The second asks whether the net transformation satisfies the event conditions.

Recent guarded realization semantics makes pathwise domains and occurrence-sensitive structure explicit in a significantly richer categorical setting [20]. The present use of a common witness is therefore best understood as a minimal realizability condition, not as a claim to originate path-sensitive semantics.

### 11.5. Scope of the closure-rank theorem

The rank theorem is deliberately restricted. It applies only when:

1. the trajectory is append-only with respect to closed coordinates;
2. each step closes at least one coordinate that was U;
3. the closure is part of strict enablement of the next step; and
4. all steps belong to one jointly realized trajectory.

Dropping these assumptions can reintroduce cycles. In particular, pairwise precedence relations supported by different witnesses need not inherit the rank argument.

### 11.6. Limitations and open directions

Several questions are intentionally left open.

First, the paper does not construct a universal event identity or null event. Functional identity is shown not to be an admissible event by default, but specialized systems may define local neutral devices under additional semantics.

Second, the paper does not impose total composition, monoid structure, group structure, or a global metric. Partiality is retained because it is already sufficient for the problem studied.

Third, the observable family is taken to be real-valued for clarity. The event-admissibility idea can be generalized to observables valued in other structures with an appropriate notion of distinguishability, but no such generalization is required for the present results.

Fourth, cross-horizon composition deserves a fuller treatment of transport coherence. Section 9 states only the conservative requirement needed here: identity and observational comparison across horizons must be declared, not presumed. No general uniqueness or composition theorem for transports is claimed.

Finally, the finite procedure in Section 10 assumes explicit representations. Symbolic or infinite-state implementations require additional machinery and are outside the present theorem scope.

## 12. Conclusion

This paper separates three levels that are often collapsed in event-based transition models: the existence of a partial composite transformation, the existence of a jointly realized path, and the constitution of an admissible composite event.

The central result shows that executable composition does not preserve event admissibility. Two individually admissible events may compose on a nonempty effective domain and yet produce a composite that is not an event because every compatible observable has zero net variation. Event status is therefore independently revalidated after composition.

Joint realizability adds a different requirement: one common witness must traverse the whole sequence. Pairwise executable compositions do not guarantee such a witness. Once a realized chain exists, compatible observable differences telescope, but the resulting net variation may still vanish and thereby defeat composite event admissibility.

In the ternary regime, U denotes traceably established non-closure rather than unknown, transition, probability, or an inconclusive verdict. Under append-only persistence, closed coordinates do not reopen within the same trajectory. This yields a finite decreasing rank for strict U-enablement chains and excludes cycles in that restricted regime.

The resulting framework is intentionally weak in algebraic commitments. It requires neither total composition nor a global identity event, and it introduces neither metric time nor probability as primitives. Its main contribution is the constitutive separation between **what can be executed** and **what qualifies as an event after execution**.


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## CRediT authorship contribution statement

Juan Antonio Lloret Egea: Conceptualization; Formal analysis; Investigation; Methodology; Validation; Writing - original draft; Writing - review and editing.

## Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

OpenAI ChatGPT (GPT-5.6 Sol) was used as a research-support tool for literature discovery, critical consistency checking, manuscript structuring, assistance with mathematical formalization and typesetting support, and English-language editing. Grok 4.5 (xAI) was used as a research-support tool for critical review of successive manuscript versions, identification of structural and mathematical improvements, and suggestions on presentation and positioning. DeepSeek-V4-Pro (DeepSeek AI) contributed to the critical review and formal consistency checking of the manuscript. All AI outputs were treated as non-authoritative research inputs. The author established, reviewed and approved all definitions, mathematical claims, proofs, interpretations and conclusions and takes full responsibility for the submitted manuscript.

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